



Semester End Project

Simple Banking System

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Programming

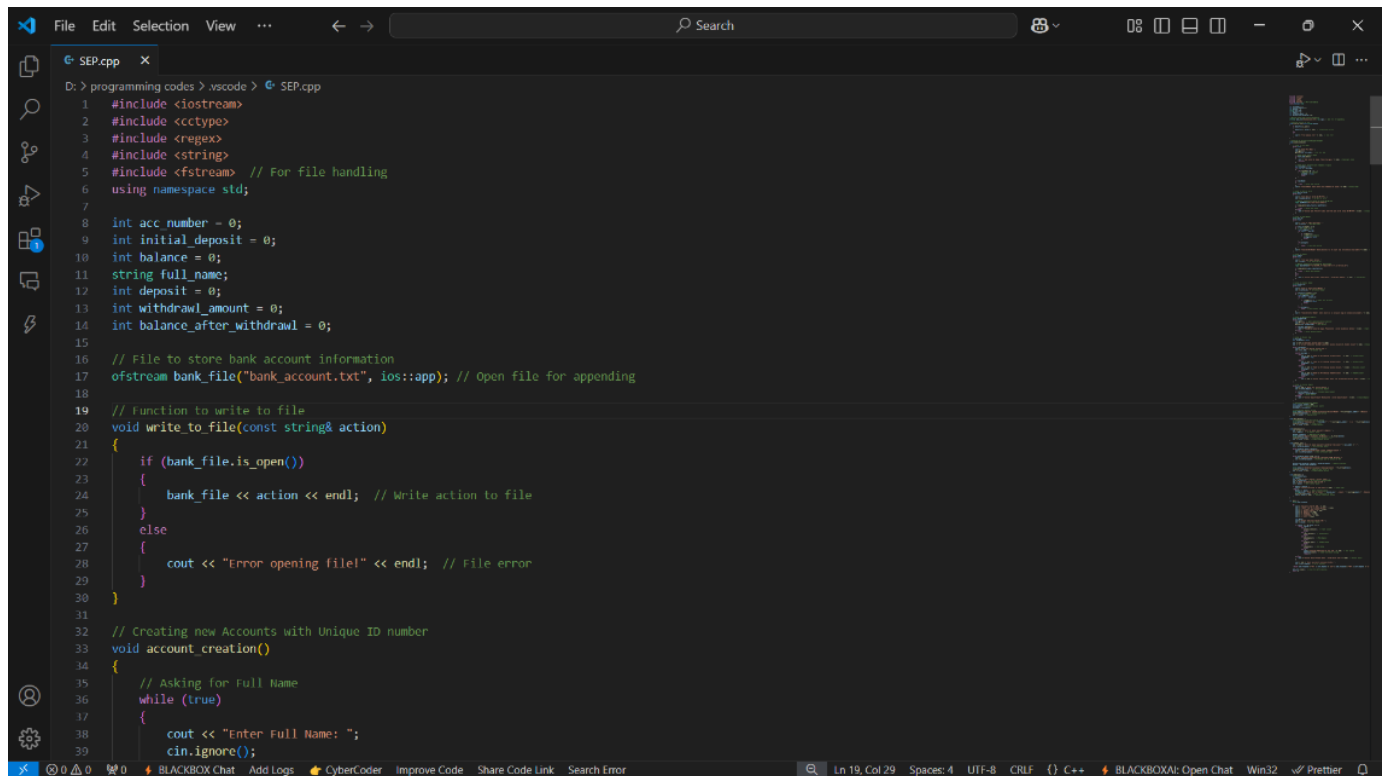
Fundamentals (Lab)

Mr. Waseem Younas



Code:

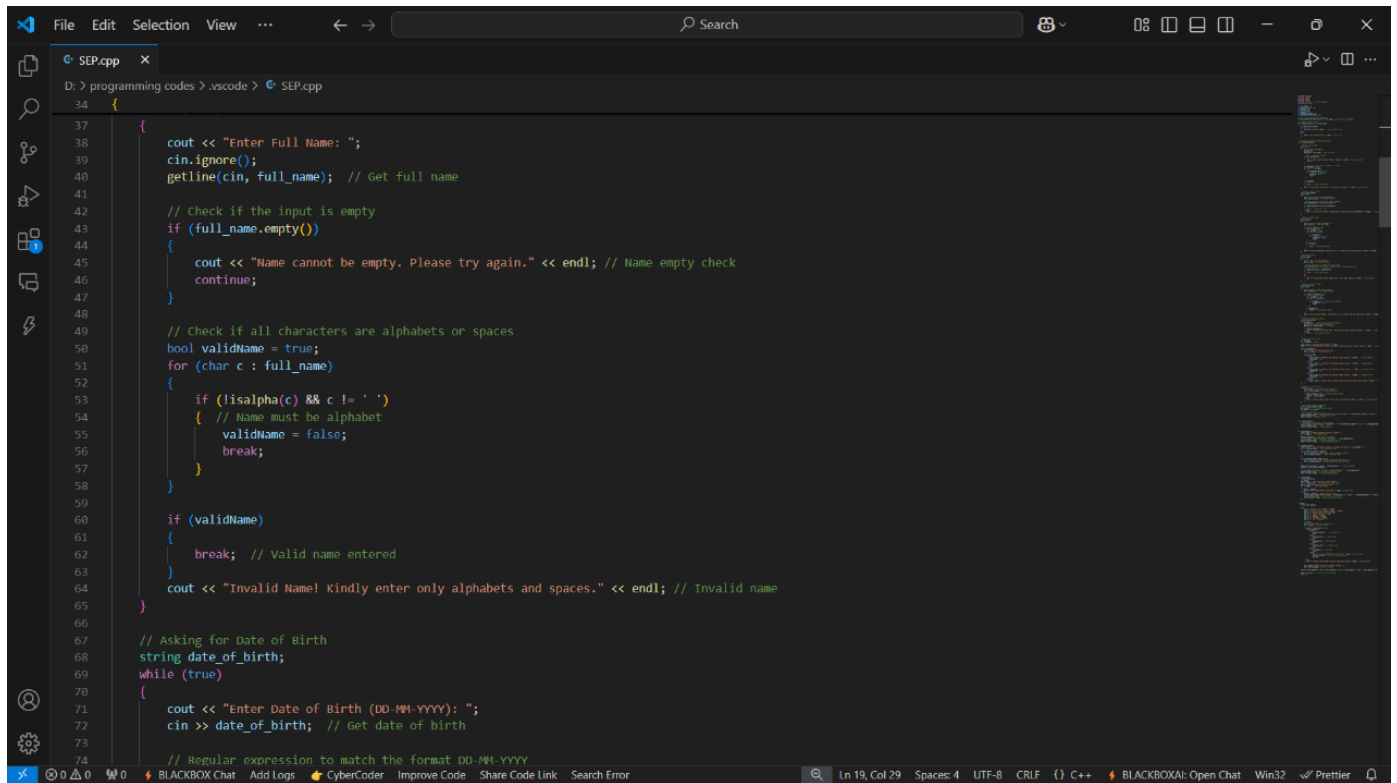
Line 1-39:



```
1 #include <iostream>
2 #include <ctype>
3 #include <regex>
4 #include <string>
5 #include <fstream> // For file handling
6 using namespace std;
7
8 int acc_number = 0;
9 int initial_deposit = 0;
10 int balance = 0;
11 string full_name;
12 int deposit = 0;
13 int withdrawl_amount = 0;
14 int balance_after_withdrawl = 0;
15
16 // File to store bank account information
17 ofstream bank_file("bank_account.txt", ios::app); // Open file for appending
18
19 // Function to write to file
20 void write_to_file(const string& action)
21 {
22     if (bank_file.is_open())
23     {
24         bank_file << action << endl; // Write action to file
25     }
26     else
27     {
28         cout << "Error opening file!" << endl; // File error
29     }
30 }
31
32 // Creating new Accounts with Unique ID number
33 void account_creation()
34 {
35     // Asking for full Name
36     while (true)
37     {
38         cout << "Enter Full Name: ";
39         cin.ignore();
```

- First we declare the required global variables and make the file in which we want to save the output.
- Then we create a function for writing some action to the file, this function writes the required function in the “text file”.

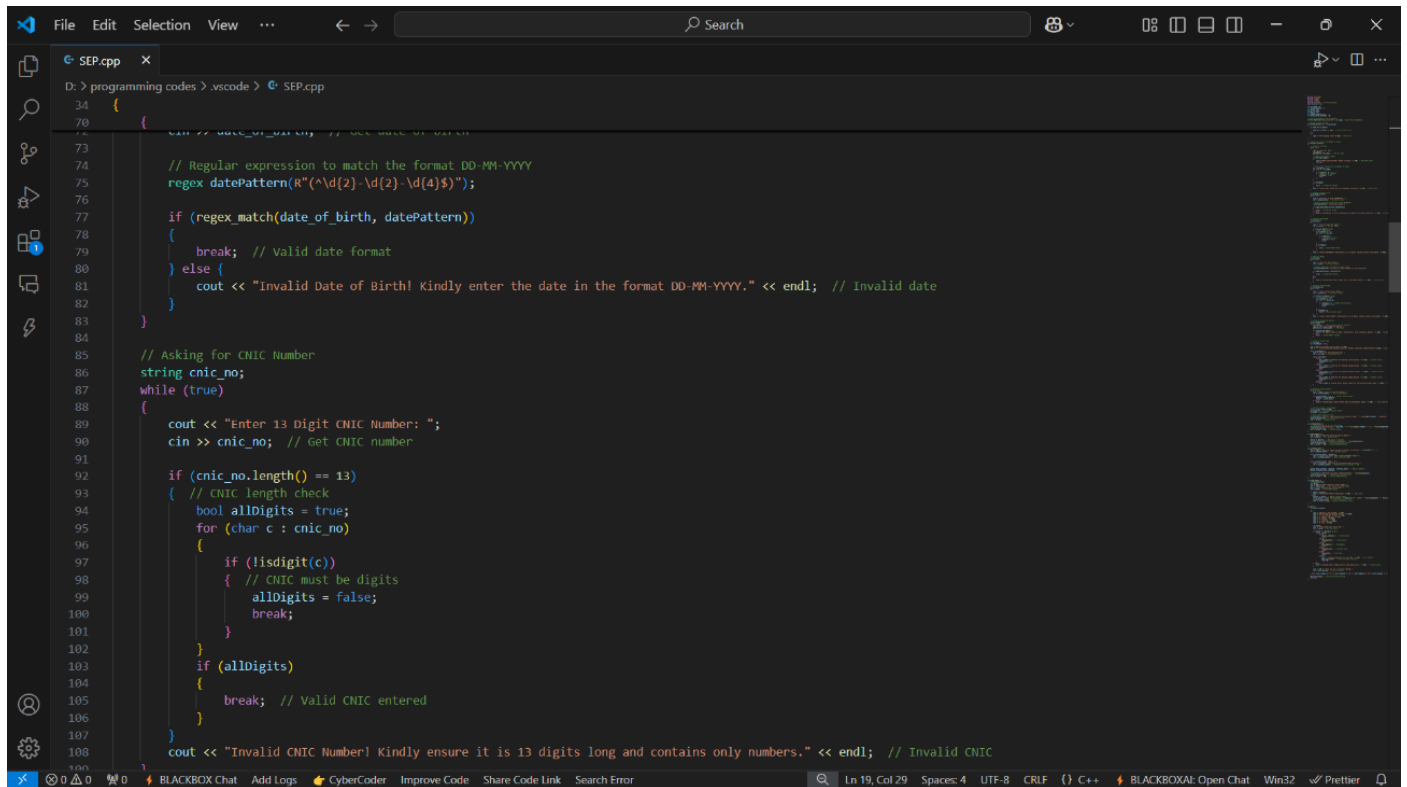
Line 39-74:



```
34 {
35
36
37     cout << "Enter Full Name: ";
38     cin.ignore();
39     getline(cin, full_name); // Get full name
40
41     // check if the input is empty
42     if (full_name.empty())
43     {
44         cout << "Name cannot be empty. Please try again." << endl; // Name empty check
45         continue;
46     }
47
48     // check if all characters are alphabets or spaces
49     bool validName = true;
50     for (char c : full_name)
51     {
52         if (!isalpha(c) && c != ' ')
53         {
54             // Name must be alphabet
55             validName = false;
56             break;
57         }
58     }
59
60     if (validName)
61     {
62         break; // Valid name entered
63     }
64     cout << "Invalid Name! Kindly enter only alphabets and spaces." << endl; // Invalid name
65 }
66
67 // Asking for Date of Birth
68 string date_of_birth;
69 while (true)
70 {
71     cout << "Enter Date of Birth (DD-MM-YYYY): ";
72     cin >> date_of_birth; // Get date of birth
73
74     // Regular expression to match the format DD-MM-YYYY
```

- Then make a new function to perform some action like account creation.
- For account creation ask the user to enter their full name and also at the same time check do input validation of the user's name.

Line 74-108:



```
73  
74 // Regular expression to match the format DD-MM-YYYY  
75 regex datePattern(R"(^\\d{2}-\\d{2}-\\d{4}$)");  
76  
77 if (regex_match(date_of_birth, datePattern))  
78 {  
79     break; // Valid date format  
80 } else {  
81     cout << "Invalid Date of Birth! Kindly enter the date in the format DD-MM-YYYY." << endl; // Invalid date  
82 }  
83 }  
84  
85 // Asking for CNIC Number  
86 string cnic_no;  
87 while (true)  
88 {  
89     cout << "Enter 13 Digit CNIC Number: ";  
90     cin >> cnic_no; // Get CNIC number  
91  
92     if (cnic_no.length() == 13)  
93     { // CNIC length check  
94         bool allDigits = true;  
95         for (char c : cnic_no)  
96         {  
97             if (!isdigit(c))  
98             { // CNIC must be digits  
99                 allDigits = false;  
100                 break;  
101             }  
102         }  
103         if (allDigits)  
104         {  
105             break; // Valid CNIC entered  
106         }  
107     }  
108     cout << "Invalid CNIC Number! Kindly ensure it is 13 digits long and contains only numbers." << endl; // Invalid CNIC
```

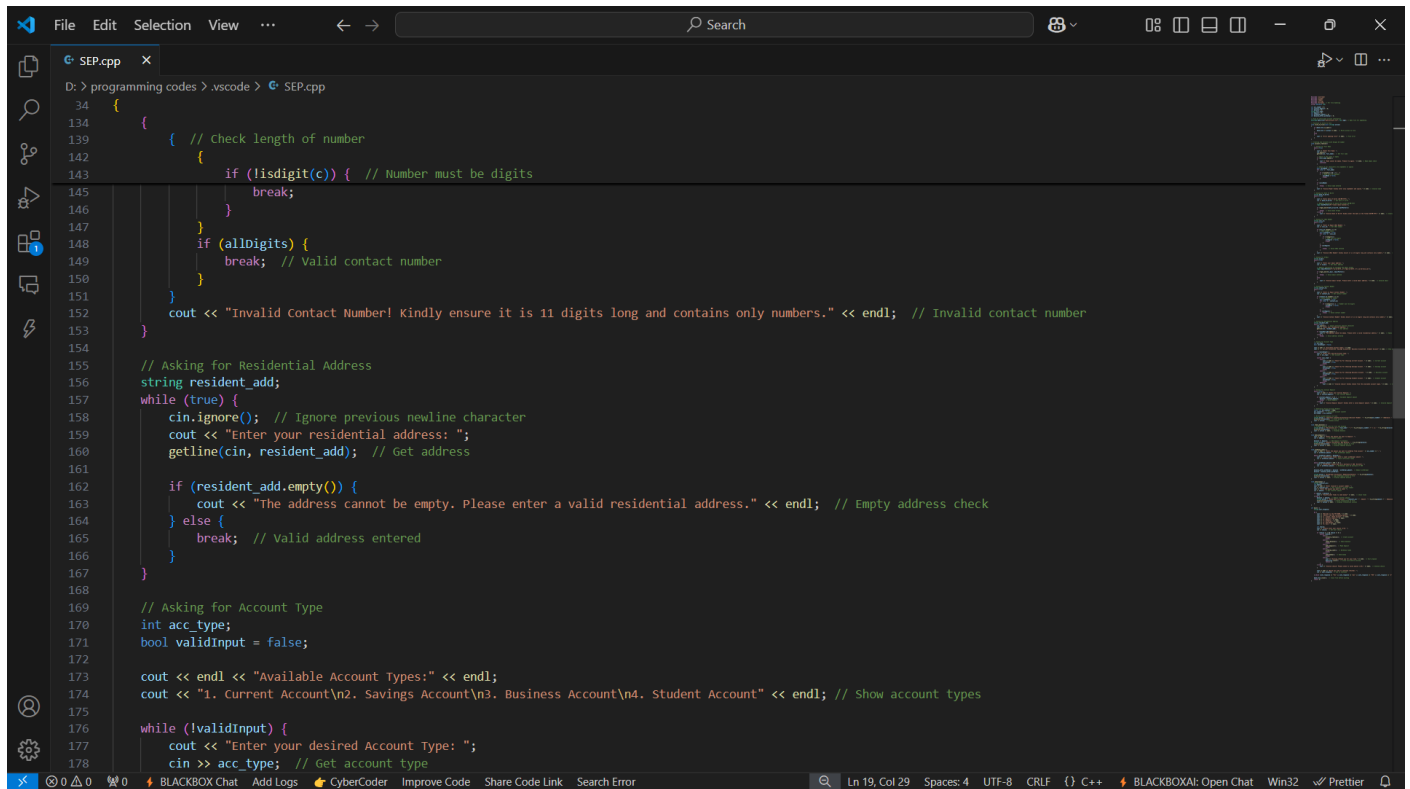
- In the next function take the CNIC Number (Citizen National Identity Card Number) from the user.
- Do input validation on the input CNIC number.

Line 108-143:

```
SEP.cpp
D:\> programming codes > .vscode > SEP.cpp
34 {
88 {
107 }
108 cout << "Invalid CNIC Number! Kindly ensure it is 13 digits long and contains only numbers." << endl; // Invalid CNIC
109 }
110
111 // Asking for E-Mail
112 string e_mail;
113 while (true)
114 {
115     cout << "Enter your email address: ";
116     cin >> e_mail; // Get email address
117
118     // Regular expression to validate the email format
119     regex emailPattern(R"([a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,})");
120
121     if (regex_match(e_mail, emailPattern))
122     {
123         break; // Valid email entered
124     }
125     else
126     {
127         cout << "Invalid email format. Please enter a valid email address." << endl; // Invalid email
128     }
129 }
130
131 // Asking for Contact Number
132 string contact_no;
133 while (true)
134 {
135     cout << "Enter 11 Digit Contact Number: ";
136     cin >> contact_no; // Get contact number
137
138     if (contact_no.length() == 11)
139     {
140         // Check length of number
141         bool allDigits = true;
142         for (char c : contact_no)
143         {
144             if (!isdigit(c)) { // Number must be digits
```

- In the next function take the e-mail of the user and perform validation on it.
- Next take the contact number of the user and also perform input validation on it.

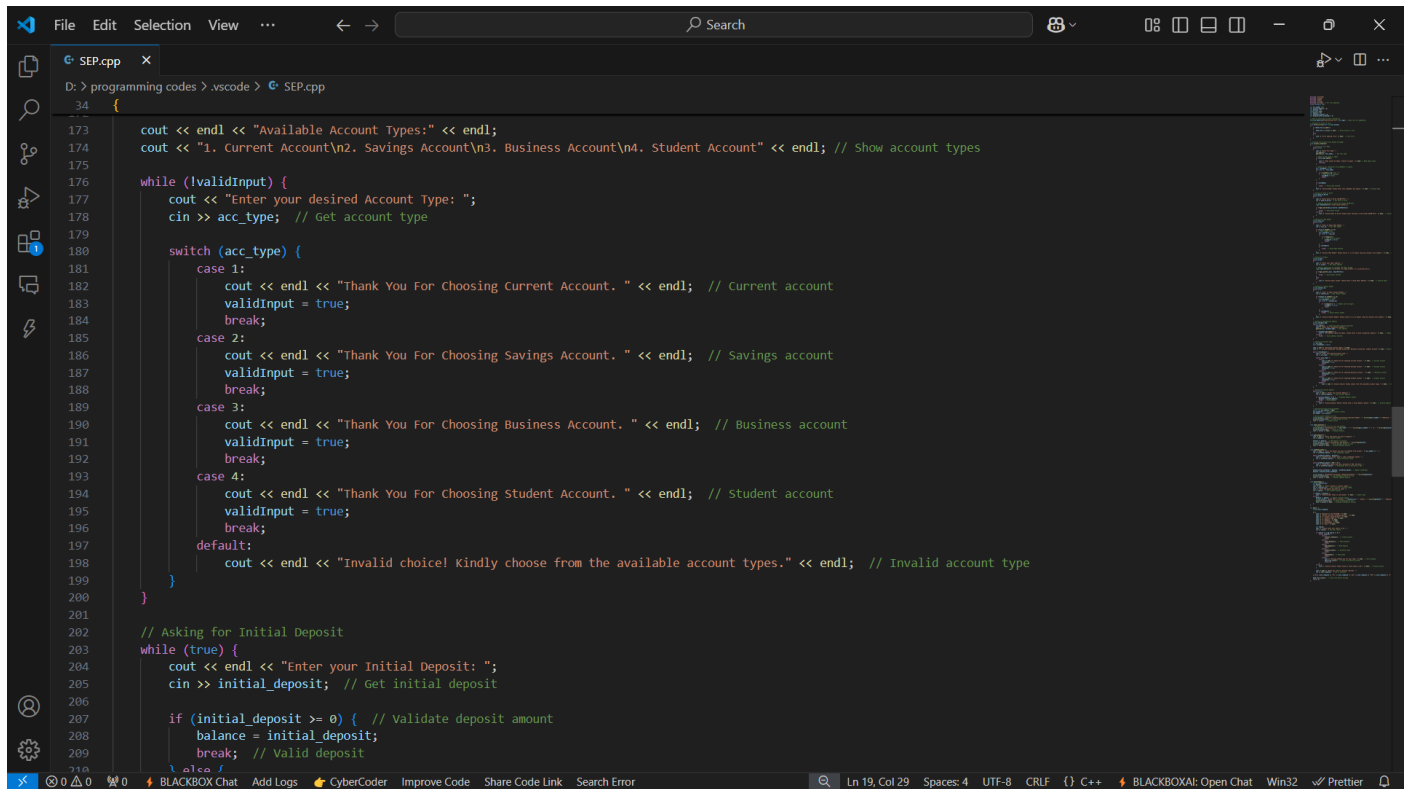
Line 143-178:



```
34 {
134 {
139 { // Check length of number
142 {
143     if (!isdigit(c)) { // Number must be digits
144         break;
145     }
146 }
147 }
148 if (allDigits) {
149     break; // Valid contact number
150 }
151 }
152 cout << "Invalid Contact Number! Kindly ensure it is 11 digits long and contains only numbers." << endl; // Invalid contact number
153 }
154
155 // Asking for Residential Address
156 string resident_add;
157 while (true) {
158     cin.ignore(); // Ignore previous newline character
159     cout << "Enter your residential address: ";
160     getline(cin, resident_add); // Get address
161
162     if (resident_add.empty()) {
163         cout << "The address cannot be empty. Please enter a valid residential address." << endl; // Empty address check
164     } else {
165         break; // Valid address entered
166     }
167 }
168
169 // Asking for Account Type
170 int acc_type;
171 bool validInput = false;
172
173 cout << endl << "Available Account Types:" << endl;
174 cout << "1. Current Account\n2. Savings Account\n3. Business Account\n4. Student Account" << endl; // Show account types
175
176 while (!validInput) {
177     cout << "Enter your desired Account Type: ";
178     cin >> acc_type; // Get account type
```

- Next take the Address of the user.
- If the user does not enter an address give a message: "The address cannot be empty. Please enter a valid residential address."
- Next ask the user what kind of bank account do they want and let the user select the desired account type.

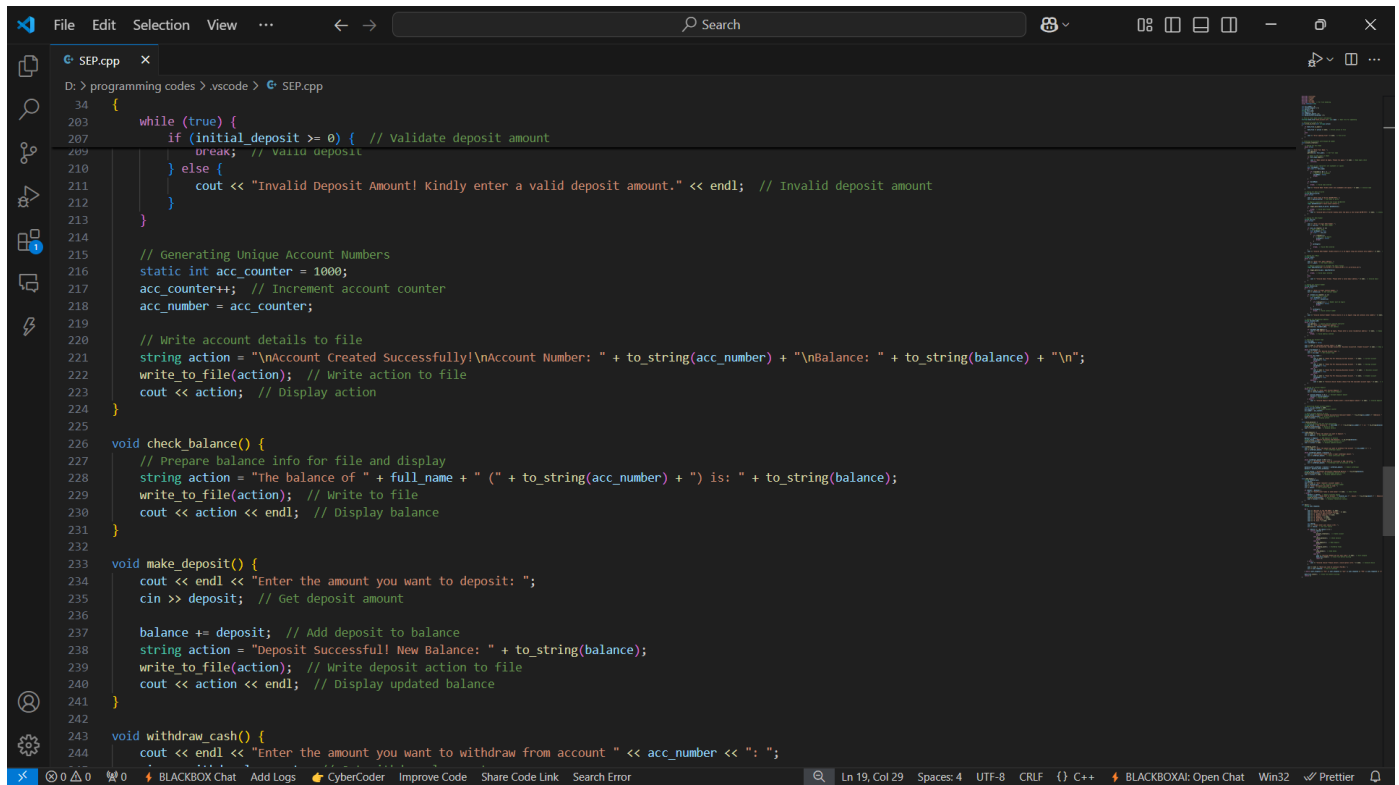
Line 178-209:



```
173     cout << endl << "Available Account Types:" << endl;
174     cout << "1. Current Account\n2. Savings Account\n3. Business Account\n4. Student Account" << endl; // Show account types
175
176     while (!validInput) {
177         cout << "Enter your desired Account Type: ";
178         cin >> acc_type; // Get account type
179
180         switch (acc_type) {
181             case 1:
182                 cout << endl << "Thank You For Choosing Current Account. " << endl; // Current account
183                 validInput = true;
184                 break;
185             case 2:
186                 cout << endl << "Thank You For Choosing Savings Account. " << endl; // Savings account
187                 validInput = true;
188                 break;
189             case 3:
190                 cout << endl << "Thank You For Choosing Business Account. " << endl; // Business account
191                 validInput = true;
192                 break;
193             case 4:
194                 cout << endl << "Thank You For Choosing Student Account. " << endl; // Student account
195                 validInput = true;
196                 break;
197             default:
198                 cout << endl << "Invalid choice! Kindly choose from the available account types." << endl; // Invalid account type
199         }
200     }
201
202     // Asking for Initial Deposit
203     while (true) {
204         cout << endl << "Enter your Initial Deposit: ";
205         cin >> initial_deposit; // Get initial deposit
206
207         if (initial_deposit >= 0) { // Validate deposit amount
208             balance = initial_deposit;
209             break; // Valid deposit
```

- After the user selects an account type prompt the user to input an initial deposit for their account.
- Note that the initial deposit must always be an amount greater than 0.

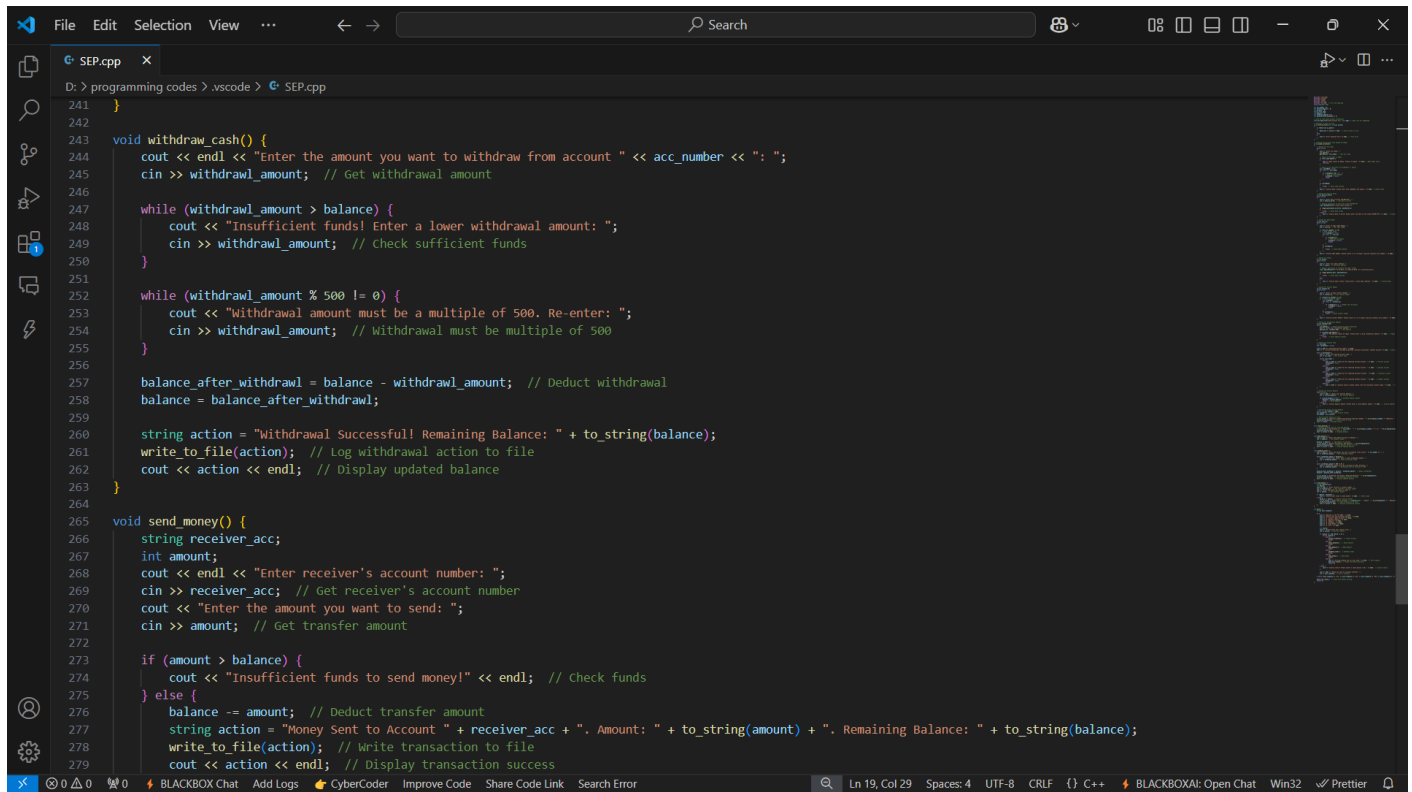
Line 210-244:



```
34 {
203     while (true) {
207         if (initial_deposit >= 0) { // Validate deposit amount
209             break; // Valid deposit
210         } else {
211             cout << "Invalid Deposit Amount! Kindly enter a valid deposit amount." << endl; // Invalid deposit amount
212         }
213     }
214
215     // Generating Unique Account Numbers
216     static int acc_counter = 1000;
217     acc_counter++; // Increment account counter
218     acc_number = acc_counter;
219
220     // Write account details to file
221     string action = "\nAccount Created Successfully!\nAccount Number: " + to_string(acc_number) + "\nBalance: " + to_string(balance) + "\n";
222     write_to_file(action); // Write action to file
223     cout << action; // Display action
224 }
225
226 void check_balance() {
227     // Prepare balance info for file and display
228     string action = "The balance of " + full_name + " (" + to_string(acc_number) + ") is: " + to_string(balance);
229     write_to_file(action); // Write to file
230     cout << action << endl; // Display balance
231 }
232
233 void make_deposit() {
234     cout << endl << "Enter the amount you want to deposit: ";
235     cin >> deposit; // Get deposit amount
236
237     balance += deposit; // Add deposit to balance
238     string action = "Deposit Successful! New Balance: " + to_string(balance);
239     write_to_file(action); // Write deposit action to file
240     cout << action << endl; // Display updated balance
241 }
242
243 void withdraw_cash() {
244     cout << endl << "Enter the amount you want to withdraw from account " << acc_number << ": ";
```

- After the account creation save the account details in a text file.
- Next check the balance of the new account (which should be equal to initial amount).
- Next make a deposit into the users account and update the balance of the user after deposit.

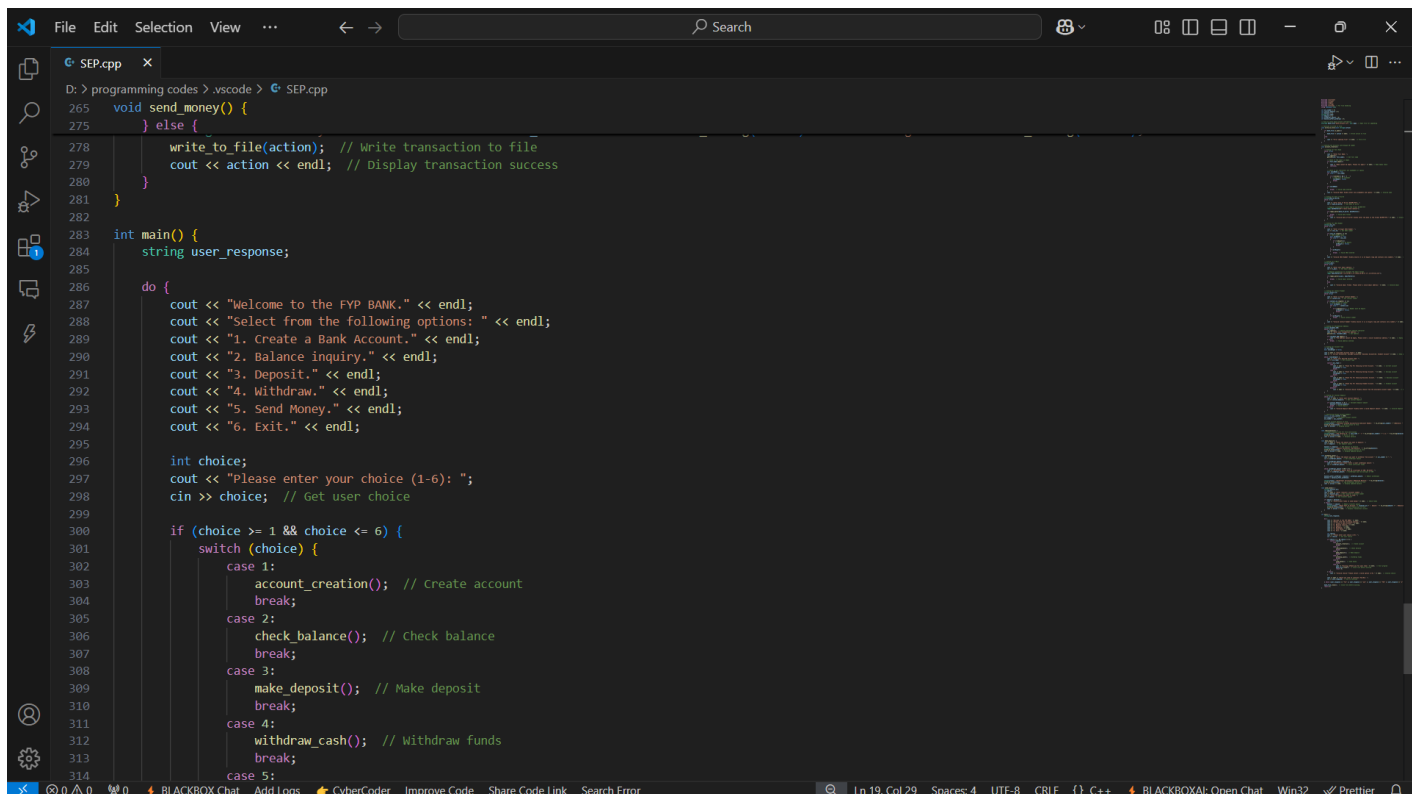
Line 245-279:



```
241 }
242
243 void withdraw_cash() {
244     cout << endl << "Enter the amount you want to withdraw from account " << acc_number << " ";
245     cin >> withdrawl_amount; // Get withdrawal amount
246
247     while (withdrawl_amount > balance) {
248         cout << "Insufficient funds! Enter a lower withdrawal amount: ";
249         cin >> withdrawl_amount; // Check sufficient funds
250     }
251
252     while (withdrawl_amount % 500 != 0) {
253         cout << "Withdrawal amount must be a multiple of 500. Re-enter: ";
254         cin >> withdrawl_amount; // Withdrawal must be multiple of 500
255     }
256
257     balance_after_withdrawl = balance - withdrawl_amount; // Deduct withdrawal
258     balance = balance_after_withdrawl;
259
260     string action = "Withdrawal Successful! Remaining Balance: " + to_string(balance);
261     write_to_file(action); // Log withdrawal action to file
262     cout << action << endl; // Display updated balance
263 }
264
265 void send_money() {
266     string receiver_acc;
267     int amount;
268     cout << endl << "Enter receiver's account number: ";
269     cin >> receiver_acc; // Get receiver's account number
270     cout << "Enter the amount you want to send: ";
271     cin >> amount; // Get transfer amount
272
273     if (amount > balance) {
274         cout << "Insufficient funds to send money!" << endl; // Check funds
275     } else {
276         balance -= amount; // Deduct transfer amount
277         string action = "Money Sent to Account " + receiver_acc + ". Amount: " + to_string(amount) + ". Remaining Balance: " + to_string(balance);
278         write_to_file(action); // Write transaction to file
279         cout << action << endl; // Display transaction success
280     }
281 }
282
283 int main() {
284     string user_response;
285
286     do {
287         cout << "Welcome to the FYP BANK." << endl;
288         cout << "Select from the following options: " << endl;
289         cout << "1. Create a Bank Account." << endl;
290         cout << "2. Balance inquiry." << endl;
291         cout << "3. Deposit." << endl;
292         cout << "4. Withdraw." << endl;
293         cout << "5. Send Money." << endl;
294         cout << "6. Exit." << endl;
295
296         int choice;
297         cout << "Please enter your choice (1-6): ";
298         cin >> choice; // Get user choice
299
300         if (choice >= 1 && choice <= 6) {
301             switch (choice) {
302                 case 1:
303                     account_creation(); // Create account
304                     break;
305                 case 2:
306                     check_balance(); // Check balance
307                     break;
308                 case 3:
309                     make_deposit(); // Make deposit
310                     break;
311                 case 4:
312                     withdraw_cash(); // Withdraw funds
313                     break;
314                 case 5:
315                     send_money(); // Send money
```

- Next make a cash withdrawal from the account and update the balance of the account.

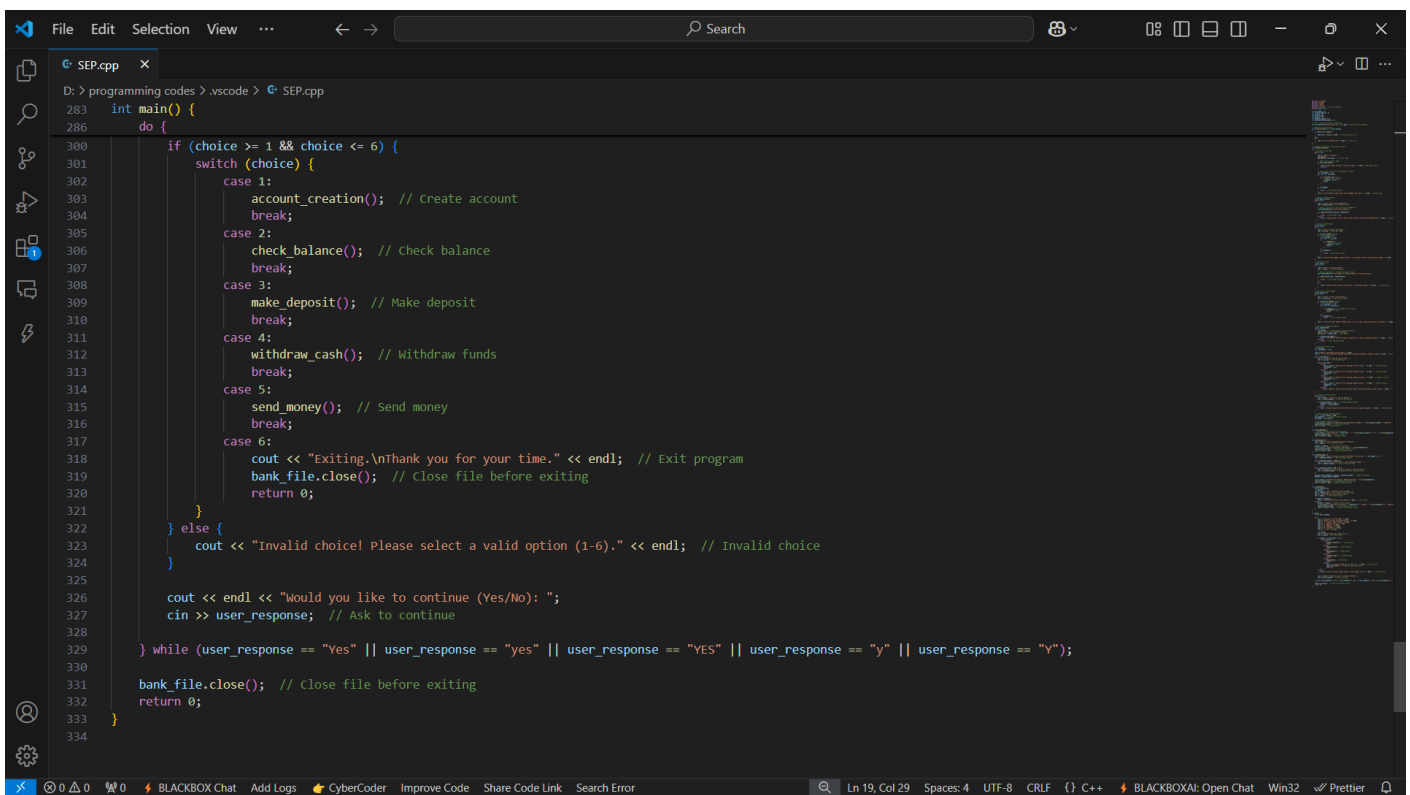
Line 280-314:



```
265 void send_money() {
275 } else {
278     write_to_file(action); // Write transaction to file
279     cout << action << endl; // Display transaction success
280 }
281 }
282
283 int main() {
284     string user_response;
285
286     do {
287         cout << "Welcome to the FYP BANK." << endl;
288         cout << "Select from the following options: " << endl;
289         cout << "1. Create a Bank Account." << endl;
290         cout << "2. Balance inquiry." << endl;
291         cout << "3. Deposit." << endl;
292         cout << "4. Withdraw." << endl;
293         cout << "5. Send Money." << endl;
294         cout << "6. Exit." << endl;
295
296         int choice;
297         cout << "Please enter your choice (1-6): ";
298         cin >> choice; // Get user choice
299
300         if (choice >= 1 && choice <= 6) {
301             switch (choice) {
302                 case 1:
303                     account_creation(); // Create account
304                     break;
305                 case 2:
306                     check_balance(); // Check balance
307                     break;
308                 case 3:
309                     make_deposit(); // Make deposit
310                     break;
311                 case 4:
312                     withdraw_cash(); // Withdraw funds
313                     break;
314                 case 5:
315                     send_money(); // Send money
```

- Next do a transaction (funds transfer) to another bank account.
- Update the balance of the account after the transaction.
- Finally create the main function and in the main function display a menu to the user for the available choices.

Line 315-333:

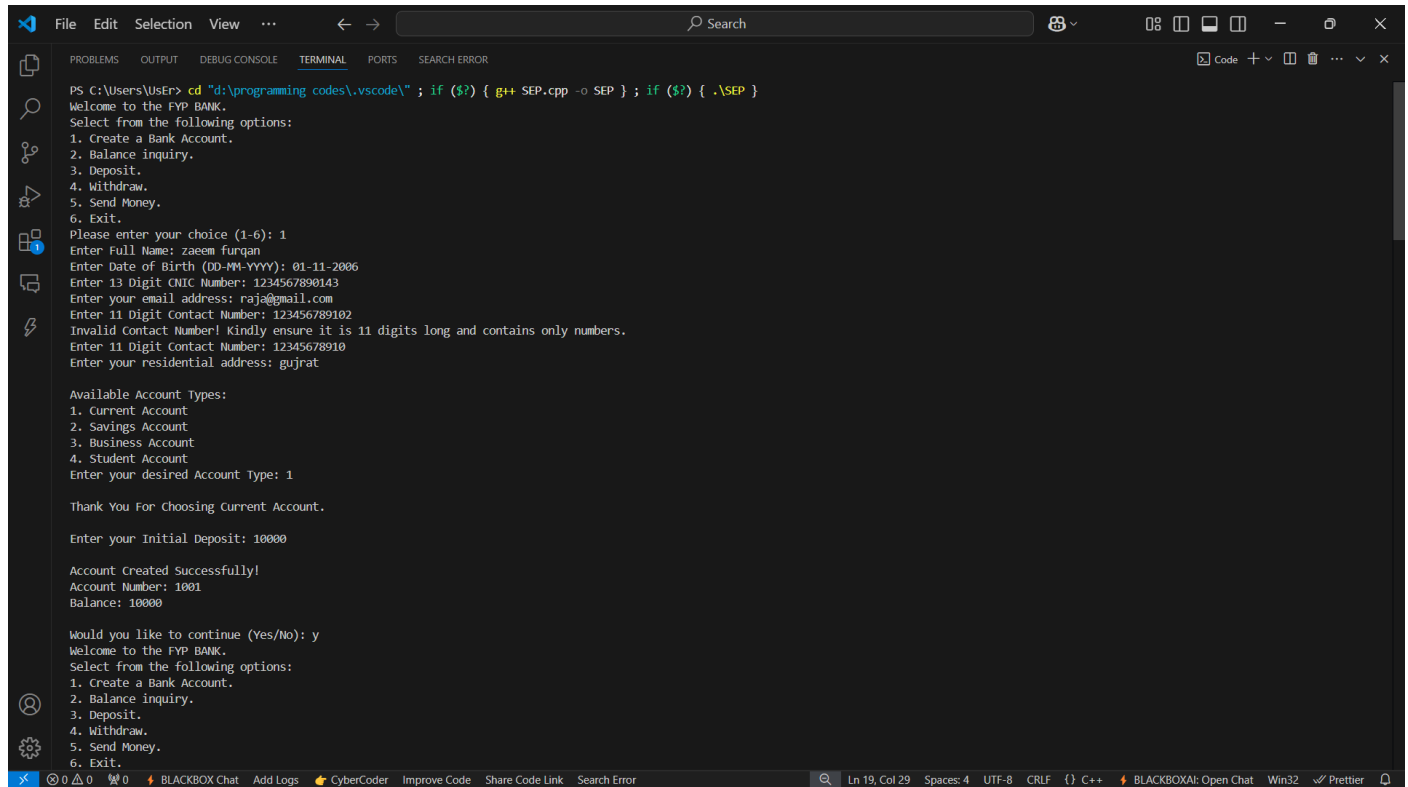


```
SEP.cpp
D:\> programming codes > .vscode > SEP.cpp
283 int main() {
286 do {
300     if (choice >= 1 && choice <= 6) {
301         switch (choice) {
302             case 1:
303                 account_creation(); // Create account
304                 break;
305             case 2:
306                 check_balance(); // Check balance
307                 break;
308             case 3:
309                 make_deposit(); // Make deposit
310                 break;
311             case 4:
312                 withdraw_cash(); // Withdraw funds
313                 break;
314             case 5:
315                 send_money(); // Send money
316                 break;
317             case 6:
318                 cout << "Exiting.\nThank you for your time." << endl; // Exit program
319                 bank_file.close(); // Close file before exiting
320                 return 0;
321         }
322     } else {
323         cout << "Invalid choice! Please select a valid option (1-6)." << endl; // Invalid choice
324     }
325
326     cout << endl << "Would you like to continue (Yes/No): ";
327     cin >> user_response; // Ask to continue
328
329     while (user_response == "Yes" || user_response == "yes" || user_response == "YES" || user_response == "y" || user_response == "Y");
330
331     bank_file.close(); // Close file before exiting
332     return 0;
333 }
```

- In the main function take a choice (1-6) from the user and use switch statement to call the necessary functions.
- Files close the text file and the program is done.

Output:

Here is a sample output of the program.



```
PS C:\Users\User> cd "d:\programming codes\.vscode\" ; if ($?) { g++ SEP.cpp -o SEP } ; if ($?) { .\SEP }
Welcome to the FYP BANK.
Select from the following options:
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
5. Send Money.
6. Exit.
Please enter your choice (1-6): 1
Enter Full Name: zaem furqan
Enter Date of Birth (DD-MM-YYYY): 01-11-2006
Enter 13 Digit CNIC Number: 1234567890143
Enter your email address: raja@gmail.com
Enter 11 Digit Contact Number: 123456789102
Invalid Contact Number! Kindly ensure it is 11 digits long and contains only numbers.
Enter 11 Digit Contact Number: 12345678910
Enter your residential address: gujrat

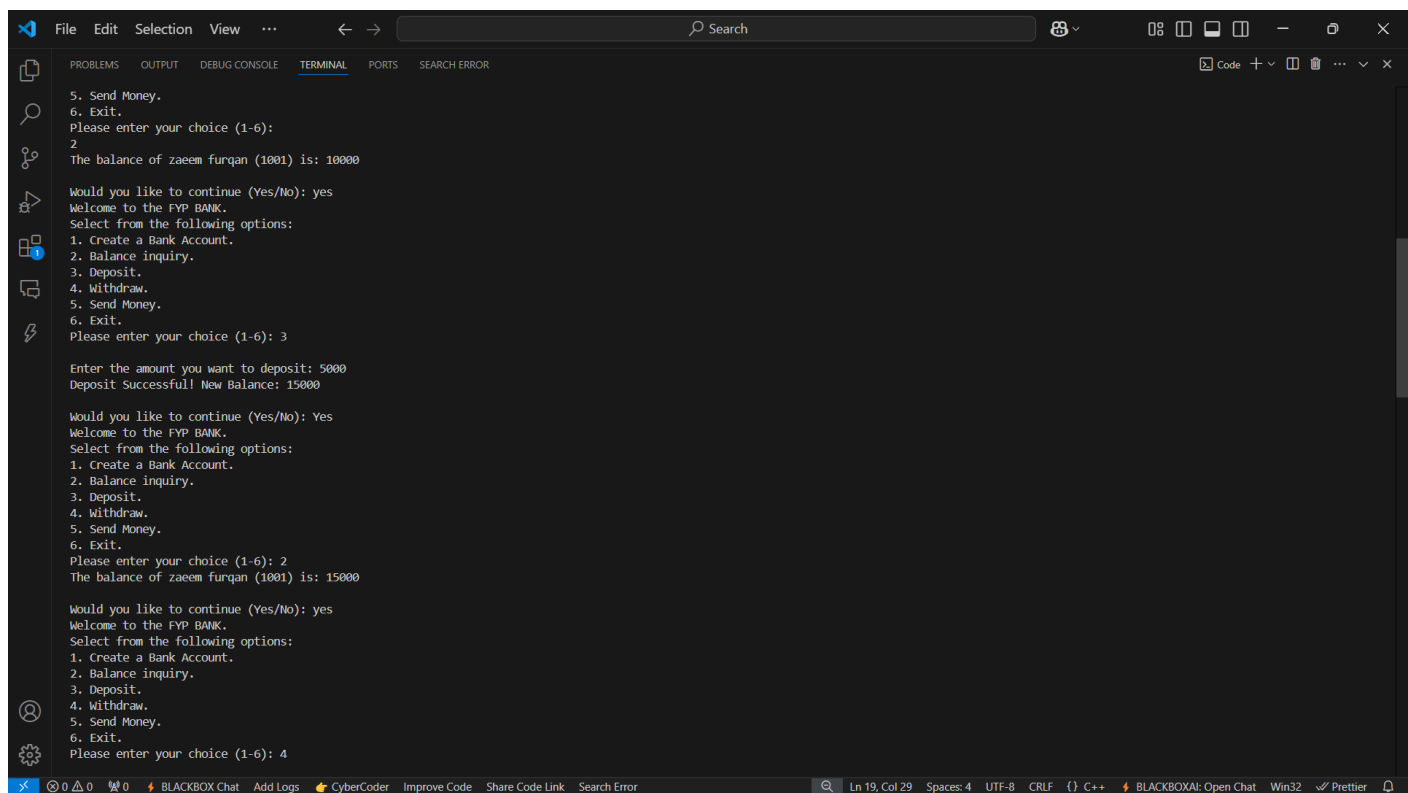
Available Account Types:
1. Current Account
2. Savings Account
3. Business Account
4. Student Account
Enter your desired Account Type: 1

Thank You For Choosing Current Account.

Enter your Initial Deposit: 10000

Account Created Successfully!
Account Number: 1001
Balance: 10000

Would you like to continue (Yes/No): y
Welcome to the FYP BANK.
Select from the following options:
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
5. Send Money.
6. Exit.
```



```
5. Send Money.
6. Exit.
Please enter your choice (1-6): 2
The balance of zaem furqan (1001) is: 10000

Would you like to continue (Yes/No): yes
Welcome to the FYP BANK.
Select from the following options:
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
5. Send Money.
6. Exit.
Please enter your choice (1-6): 3

Enter the amount you want to deposit: 5000
Deposit Successful! New Balance: 15000

Would you like to continue (Yes/No): Yes
Welcome to the FYP BANK.
Select from the following options:
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
5. Send Money.
6. Exit.
Please enter your choice (1-6): 2
The balance of zaem furqan (1001) is: 15000

Would you like to continue (Yes/No): yes
Welcome to the FYP BANK.
Select from the following options:
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
5. Send Money.
6. Exit.
Please enter your choice (1-6): 4
```

```
File Edit Selection View ... Search
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR
Please enter your choice (1-6): 4
Enter the amount you want to withdraw from account 1001: 2000
Withdrawal Successful! Remaining Balance: 13000
Would you like to continue (Yes/No): yes
Welcome to the FYP BANK.
Select from the following options:
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
5. Send Money.
6. Exit.
Please enter your choice (1-6): 5
5. Send Money.
6. Exit.
Please enter your choice (1-6): 5
6. Exit.
Please enter your choice (1-6): 5
Please enter your choice (1-6): 5

Enter receiver's account number: 1002
Enter receiver's account number: 1002
Enter the amount you want to send: 3000
Enter the amount you want to send: 3000
Money Sent to Account 1002. Amount: 3000. Remaining Balance: 10000

Would you like to continue (Yes/No): y
Would you like to continue (Yes/No): y
Welcome to the FYP BANK.
Select from the following options:
1. Create a Bank Account.
Welcome to the FYP BANK.
Select from the following options:
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
```

```
File Edit Selection View ... Search
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR
2. Balance inquiry.
3. Deposit.
1. Create a Bank Account.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
5. Send Money.
2. Balance inquiry.
3. Deposit.
4. Withdraw.
5. Send Money.
6. Exit.
Please enter your choice (1-6): 6
4. Withdraw.
5. Send Money.
6. Exit.
Please enter your choice (1-6): 6
5. Send Money.
6. Exit.
Please enter your choice (1-6): 6
Exiting.
6. Exit.
Please enter your choice (1-6): 6
Exiting.
Exiting.
Thank you for your time.
PS D:\programming codes\.\vscode>
```

Thank you for your time!

END.